

CSD18532Q5B 60V N-Channel NexFET™ Power MOSFETs

1 Features

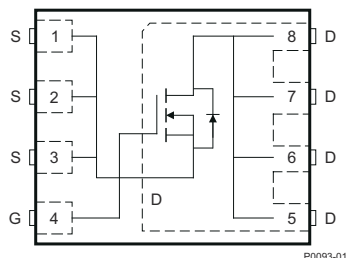
- Ultra-low Q_g and Q_{gd}
- Low thermal resistance
- Avalanche rated
- Logic level
- Lead-free terminal plating
- RoHS compliant
- Halogen free
- SON 5mm × 6mm plastic package

2 Applications

- DC-DC conversion
- Secondary side synchronous rectifier
- Isolated converter primary side switch
- Motor control

3 Description

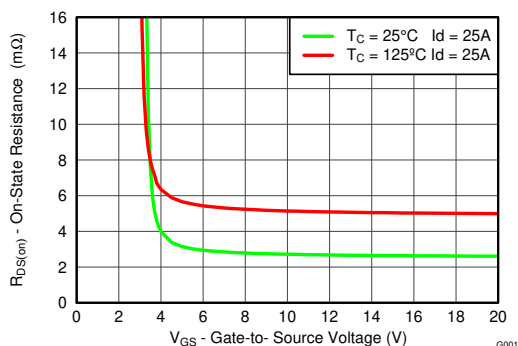
This 2.5mΩ, 60V SON 5mm × 6mm NexFET™ power MOSFET is designed to minimize losses in power conversion applications.



Top View

Product Summary

$T_A = 25^\circ\text{C}$		TYPICAL VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	60	V
Q_g	Gate Charge Total (10V)	44	nC
Q_{gd}	Gate Charge Gate-to-Drain	6.9	nC



$R_{DS(on)}$ vs V_{GS}

Product Summary (continued)

$T_A = 25^\circ\text{C}$		TYPICAL VALUE	UNIT
$R_{DS(on)}$	Drain-to-Source On-Resistance	$V_{GS} = 4.5\text{V}$	3.3
		$V_{GS} = 10\text{V}$	2.5
$V_{GS(th)}$	Threshold Voltage	1.8	V

Package Information

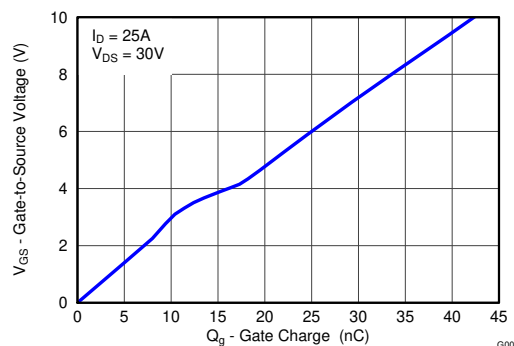
DEVICE	QTY	MEDIA	PACKAGE ^{(1) (2)}	SHIP
CSD18532Q5B	2500	13-Inch Reel	SON	Tape and Reel
CSD18532Q5BT	250	13-Inch Reel	5.00mm × 6.00mm Plastic Package	

- (1) For all available packages, see the orderable addendum at the end of the datasheet.
- (2) The package size (length × width) is a nominal value and includes pins, where applicable.

Absolute Maximum Ratings

$T_A = 25^\circ\text{C}$		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	60	V
V_{GS}	Gate-to-Source Voltage	±20	V
I_D	Continuous Drain Current (Package Limited)	100	A
	Continuous Drain Current (Silicon Limited), $T_C = 25^\circ\text{C}$	172	
	Continuous Drain Current ⁽¹⁾	23	
I_{DM}	Pulsed Drain Current ⁽²⁾	400	A
P_D	Power Dissipation ⁽¹⁾	3.2	W
	Power Dissipation, $T_C = 25^\circ\text{C}$	156	
T_J	Operating Junction Temperature	–55 to 150	$^\circ\text{C}$
T_{stg}	Storage Temperature		
E_{AS}	Avalanche Energy, Single Pulse $I_D = 80\text{A}$, $L = 0.1\text{mH}$, $R_G = 25\Omega$	320	mJ

- (1) Typical $R_{\theta JA} = 40^\circ\text{C/W}$ on a 1-in², 2oz Cu pad on a 0.06-in thick FR4 PCB.
- (2) Maximum $R_{\theta JC} = 0.8^\circ\text{C/W}$, pulse duration ≤ 100 μs, duty cycle ≤ 1%.



Gate Charge



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4 Specifications

4.1 Electrical Characteristics

$T_A = 25^\circ\text{C}$ unless otherwise stated

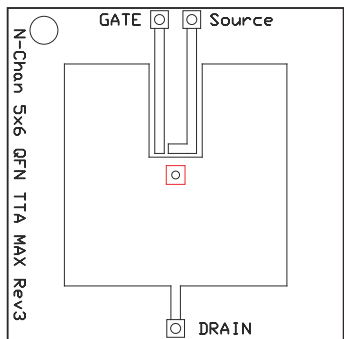
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
STATIC CHARACTERISTICS						
BV _{DSS}	Drain-to-source voltage	V _{GS} = 0V, I _D = 250μA	60			V
I _{DSS}	Drain-to-source leakage current	V _{GS} = 0V, V _{DS} = 48V			1	μA
I _{GSS}	Gate-to-source leakage current	V _{DS} = 0V, V _{GS} = 20V			100	nA
V _{GS(th)}	Gate-to-source threshold voltage	V _{DS} = V _{GS} , I _D = 250μA	1.5	1.8	2.2	V
R _{DS(on)}	Drain-to-source on-resistance	V _{GS} = 4.5V, I _D = 25A	3.3		4.3	mΩ
		V _{GS} = 10V, I _D = 25A	2.5		3.2	
g _{fs}	Transconductance	V _{DS} = 30V, I _D = 25A	143			S
DYNAMIC CHARACTERISTICS						
C _{iss}	Input capacitance	V _{GS} = 0V, V _{DS} = 30V, f = 1MHz		3900	5070	pF
C _{oss}	Output capacitance			470	611	pF
C _{rss}	Reverse transfer capacitance			13	17	pF
R _G	Series gate resistance	V _{DS} = 30V, I _D = 25A		1.2	2.4	Ω
Q _g	Gate charge total (10V)			44	58	nC
Q _{gd}	Gate charge gate-to-drain			6.9		nC
Q _{gs}	Gate charge gate-to-source			10		nC
Q _{g(th)}	Gate charge at V _{th}			6.3		nC
Q _{oss}	Output charge	V _{DS} = 30V, V _{GS} = 0V		52		nC
t _{d(on)}	Turnon delay time	V _{DS} = 30V, V _{GS} = 10V, I _{DS} = 25A, R _G = 0 Ω		5.8		ns
t _r	Rise time			7.2		ns
t _{d(off)}	Turnoff delay time			22		ns
t _f	Fall time			3.1		ns
DIODE CHARACTERISTICS						
V _{SD}	Diode forward voltage	I _{SD} = 25A, V _{GS} = 0V		0.8	1	V
Q _{rr}	Reverse recovery charge	V _{DS} = 30V, I _F = 25A, di/dt = 300A/μs		111		nC
t _{rr}	Reverse recovery time			49		ns

4.2 Thermal Information

$T_A = 25^\circ\text{C}$ unless otherwise stated

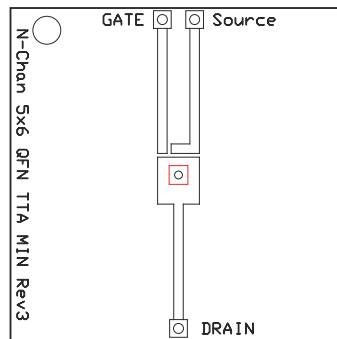
THERMAL METRIC		MIN	TYP	MAX	UNIT
$R_{\theta JC}$	Junction-to-case thermal resistance ⁽¹⁾			0.8	$^\circ\text{C/W}$
$R_{\theta JA}$	Junction-to-ambient thermal resistance ^{(1) (2)}			50	$^\circ\text{C/W}$

- (1) $R_{\theta JC}$ is determined with the device mounted on a 1in² (6.45cm²), 2oz. (0.071mm) thick Cu pad on a 1.5in × 1.5in (3.81cm × 3.81cm), 0.06in (1.52mm) thick FR4 PCB. $R_{\theta JC}$ is specified by design, whereas $R_{\theta JA}$ is determined by the user's board design.
- (2) Device mounted on FR4 material with 1in² (6.45cm²), 2oz (0.071mm) thick Cu.



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Maximum $R_{\theta JA} = 50^{\circ}\text{C/W}$
when mounted on 1 in²
(6.45cm²) of 2oz (0.071mm)
thick Cu.

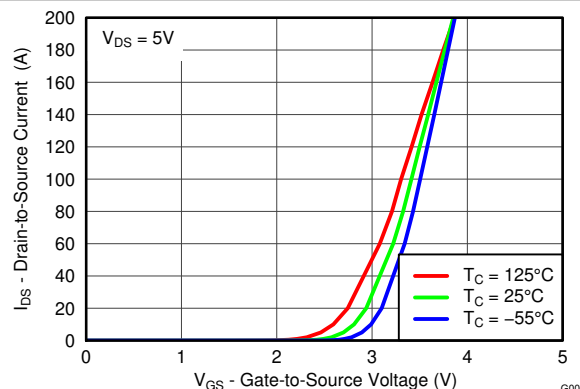
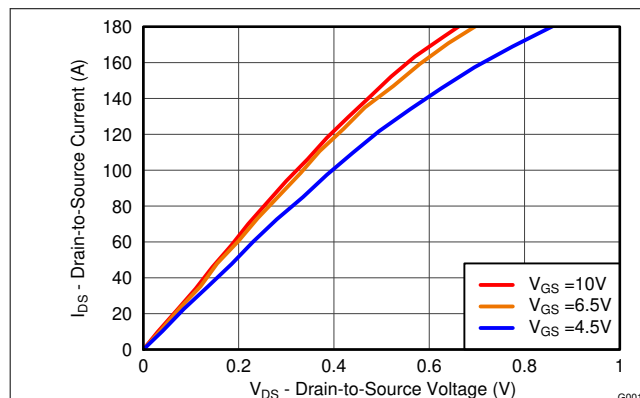
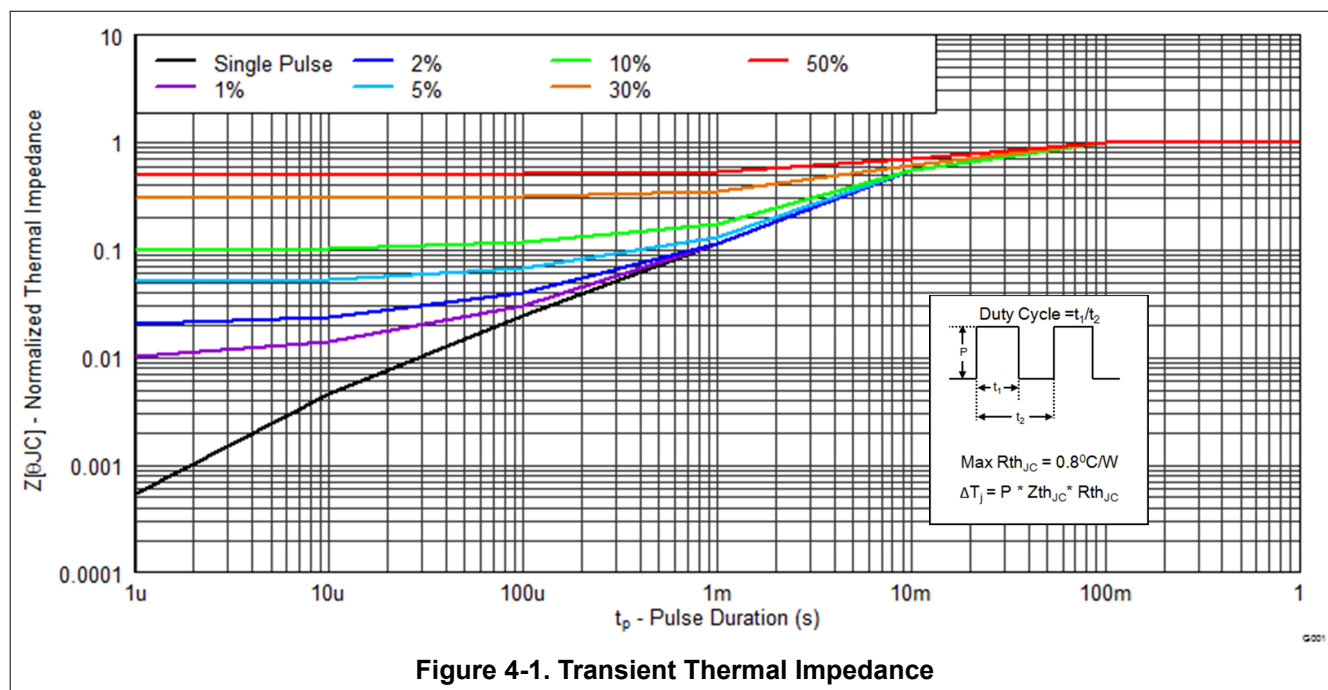


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Maximum $R_{\theta JA} = 125^{\circ}\text{C/W}$
when mounted on a
minimum pad area of 2oz
(0.071mm) thick Cu.

4.3 Typical MOSFET Characteristics

$T_A = 25^{\circ}\text{C}$ unless otherwise stated



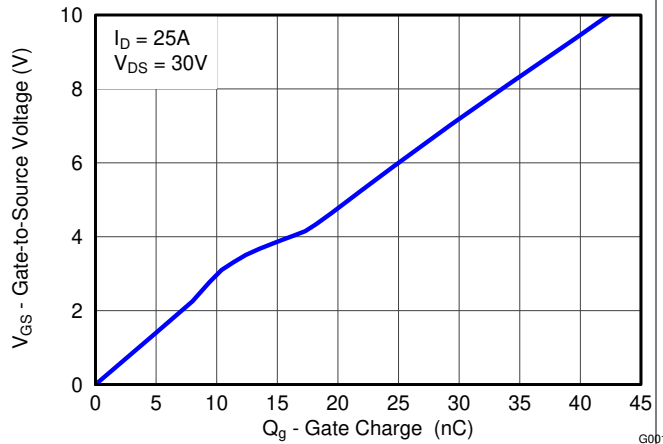


Figure 4-4. Gate Charge

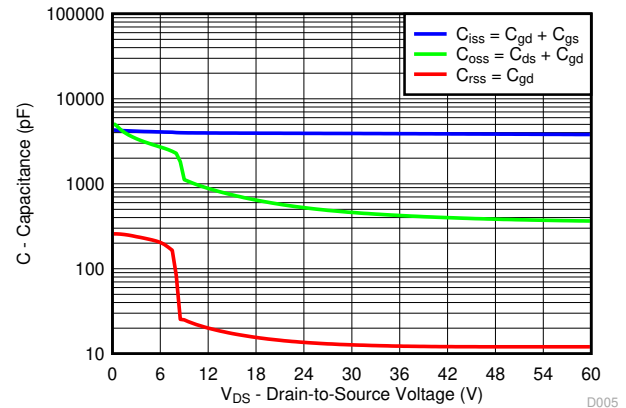


Figure 4-5. Capacitance

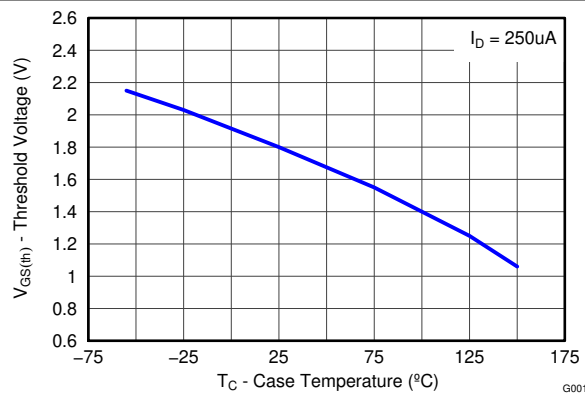


Figure 4-6. Threshold Voltage vs Temperature

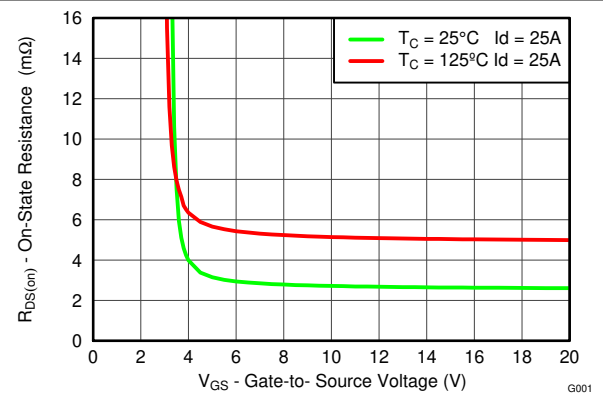


Figure 4-7. On-State Resistance vs Gate-to-Source Voltage

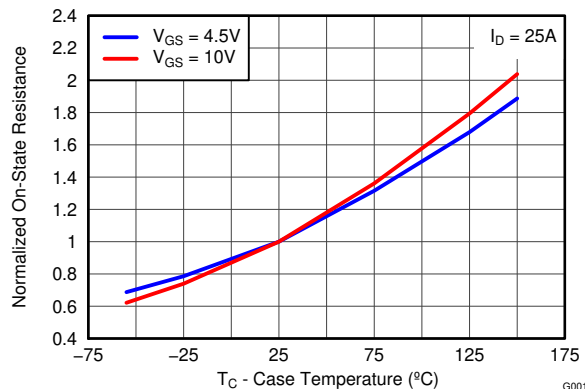


Figure 4-8. Normalized On-State Resistance vs Temperature

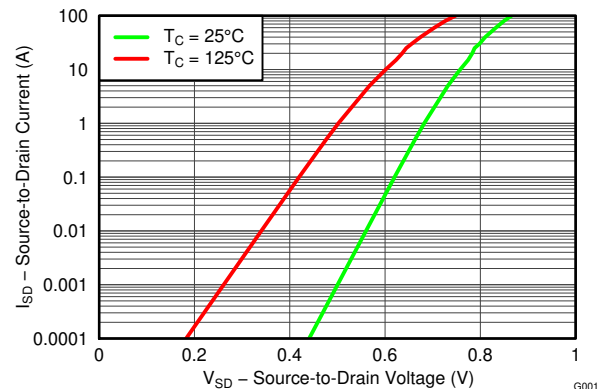


Figure 4-9. Typical Diode Forward Voltage

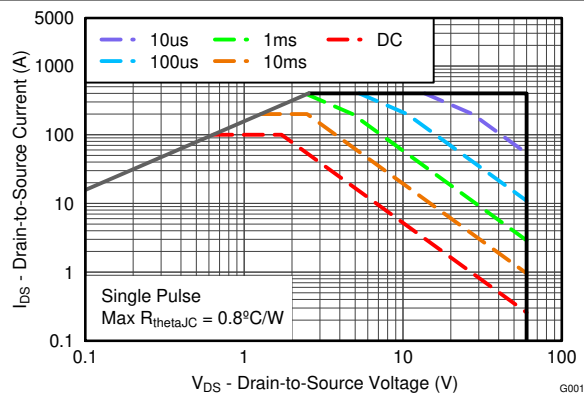


Figure 4-10. Maximum Safe Operating Area

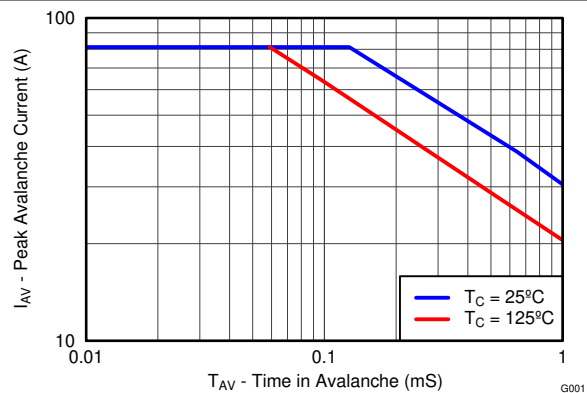


Figure 4-11. Single Pulse Unclamped Inductive Switching

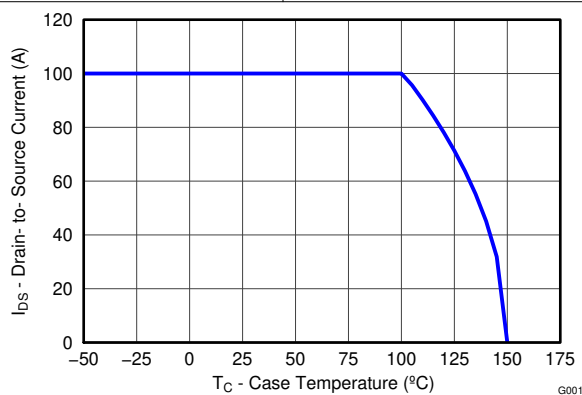


Figure 4-12. Maximum Drain Current vs Temperature

5 Device and Documentation Support

5.1 Third-Party Products Disclaimer

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5.2 Documentation Support

5.2.1 Related Documentation

5.3 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

5.4 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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5.6 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

5.7 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

6 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision D (February 2018) to Revision E (July 2026)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document.....	1

Changes from Revision C (May 2017) to Revision D (February 2018)	Page
• Extended the V_{DS} on Figure 4-5 to 60V.....	4

7 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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